Insights through
avanced statistics

Understanding Drivers of Diabetes Risk: From Prediction to Actionable Insights

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Diabetes is a very complex condition that is affected by many factors including lifestyle and genetics. It has been coined an epidemic of the twenty-first century.

Can we use advanced statistics to identify some significant lifestyle factors and their interaction to create actionable steps to reduce the risk of diabetes?

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The Dataset

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The dataset includes features from following domains:

- Comorbidity - exclude, not drivers
- Food consumption - fruit and veg
- Physical activity - In the past 30 days
- Healthcare and GP visits - Healthcare insurance, no GP visit due to high cost
- Demographics and socio economic status - Age, gender, income, education.
- Note: Age is encoded in 1 - (18-24), 2 - (25-29)....13-(80 or older)
- Smoking and Alcohol - smoked 100 cigarettes in entire life, heavy drinker
- BMI and other metrics

Results of Dataset inspection: Identified 14 primary drivers.

The final table has 50736 rows and 14 predictive features with 13% positive/86% negative class imbalance

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Model Selection

This is a classification problem with class imbalance, Large number of columns is binary. Two binary classifiers were selected and performance compared in form of precision/recall.

Logistic Regression

Pros: Fits into linear relationship, High explainability

Cons: Does not catupre non linear relationships between featues and target

XGBoost

Pros: very robust, captures non-lenear relationships

Cons:Low explainability have to use additional techniques to explain results

Owards to results and discussion

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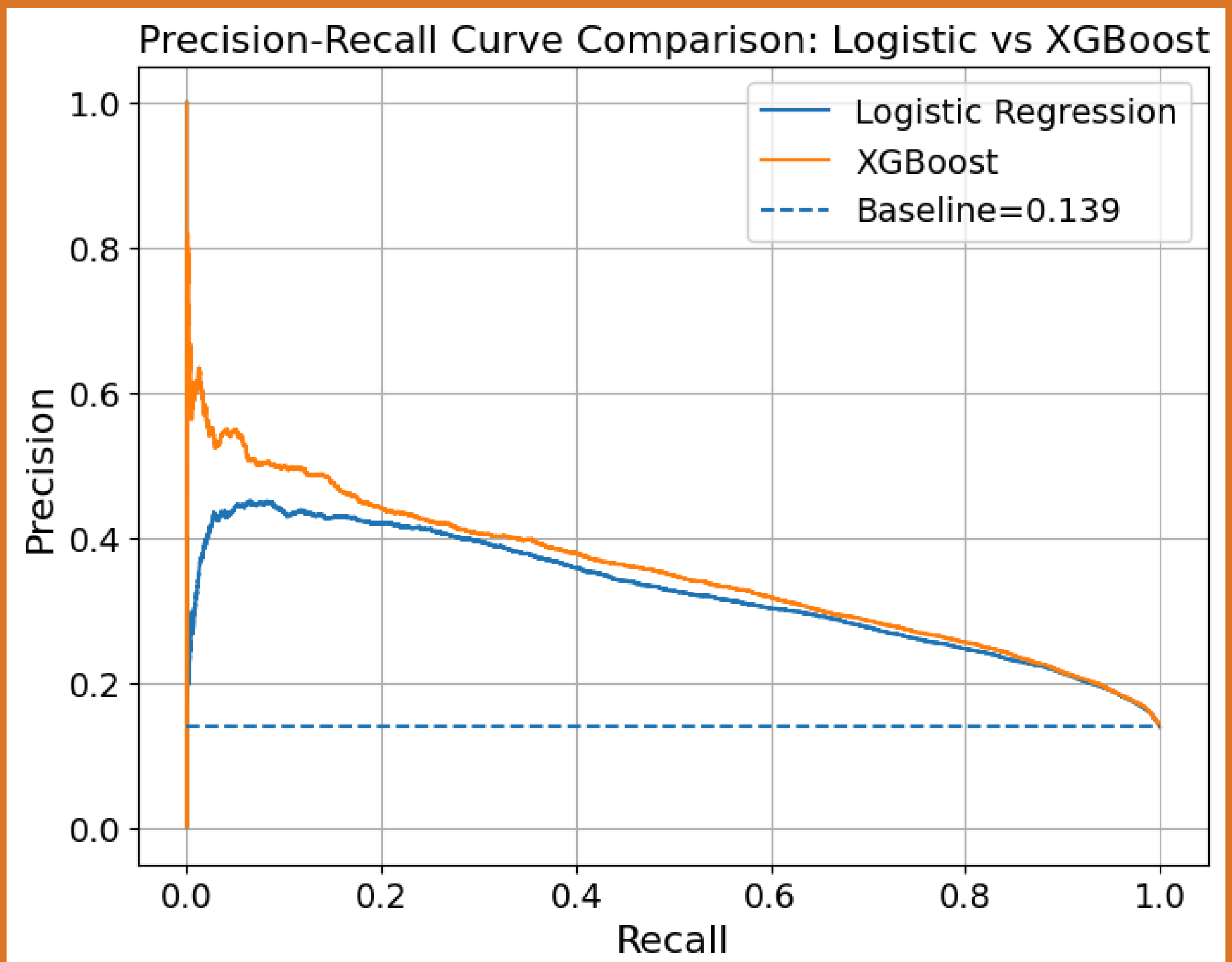

XGBoost with hyperparameter tunning performs marginally better then logistic regression.In a group of the top 20% (20% recall) just over 40% actually have diabetes.This is true for both algorithms. This is also significantly better then 14% precision for coin toss

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Given similar performance of the two algorithms logistic regression is chosen, since it allows for greater explainability

Trained Logistic Regression algorithm can be used now to identify how different drivers affect the outcome of the disease. Complex interactions can also be explored

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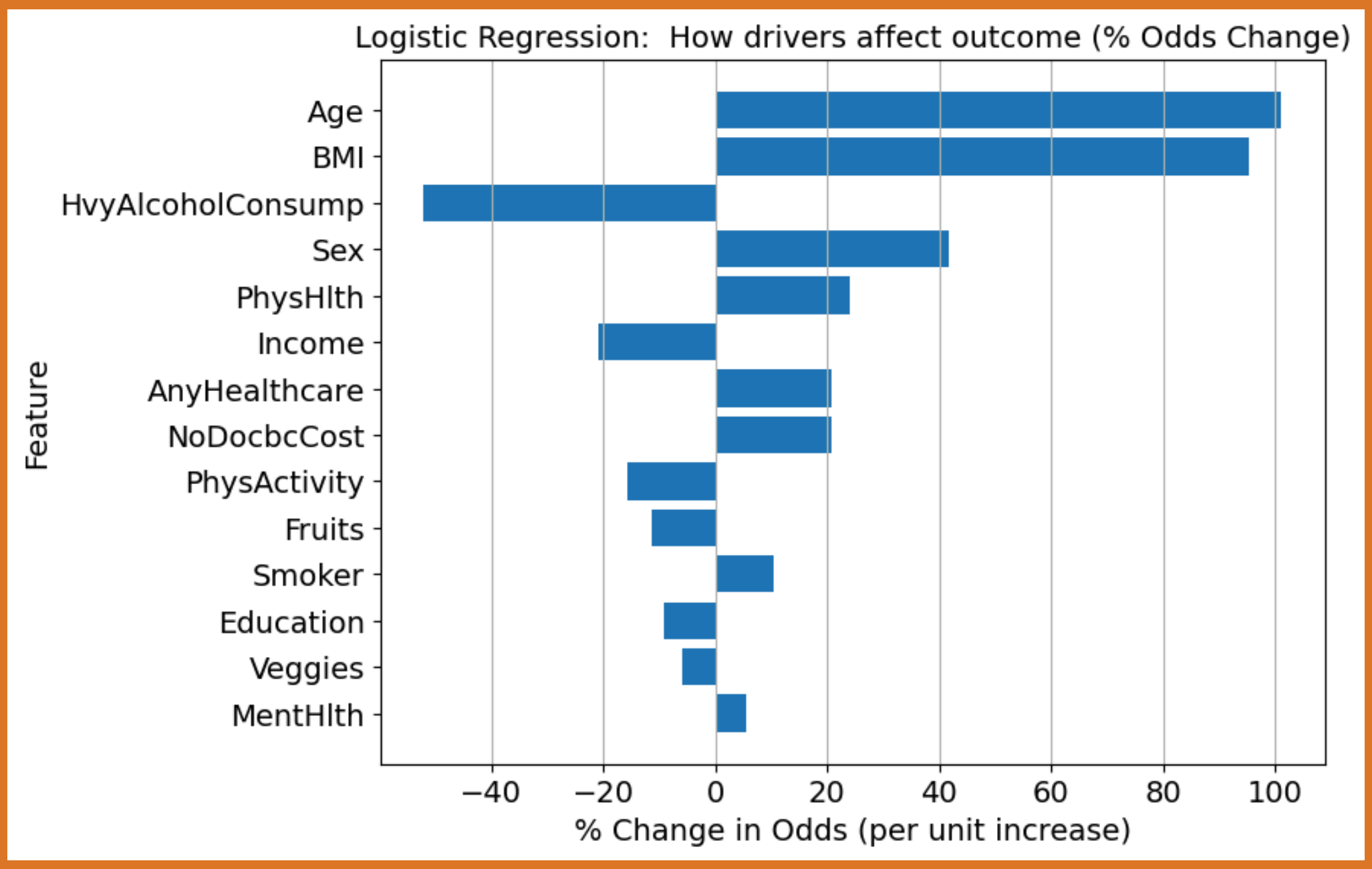

Above figure shows the odds of change in the decision of algorithm (positive if diabetes, negative absence of diabetes) in the x-axis: change in odds if individual predictor is increased by a single unit, while rest of the drivers are constant.

Age and BMI are the two strongest drivers, that can cause up to 2 time increase in likelihood of diabetes. Heavy alcohold consumption surprisingly reduces diabetes chance

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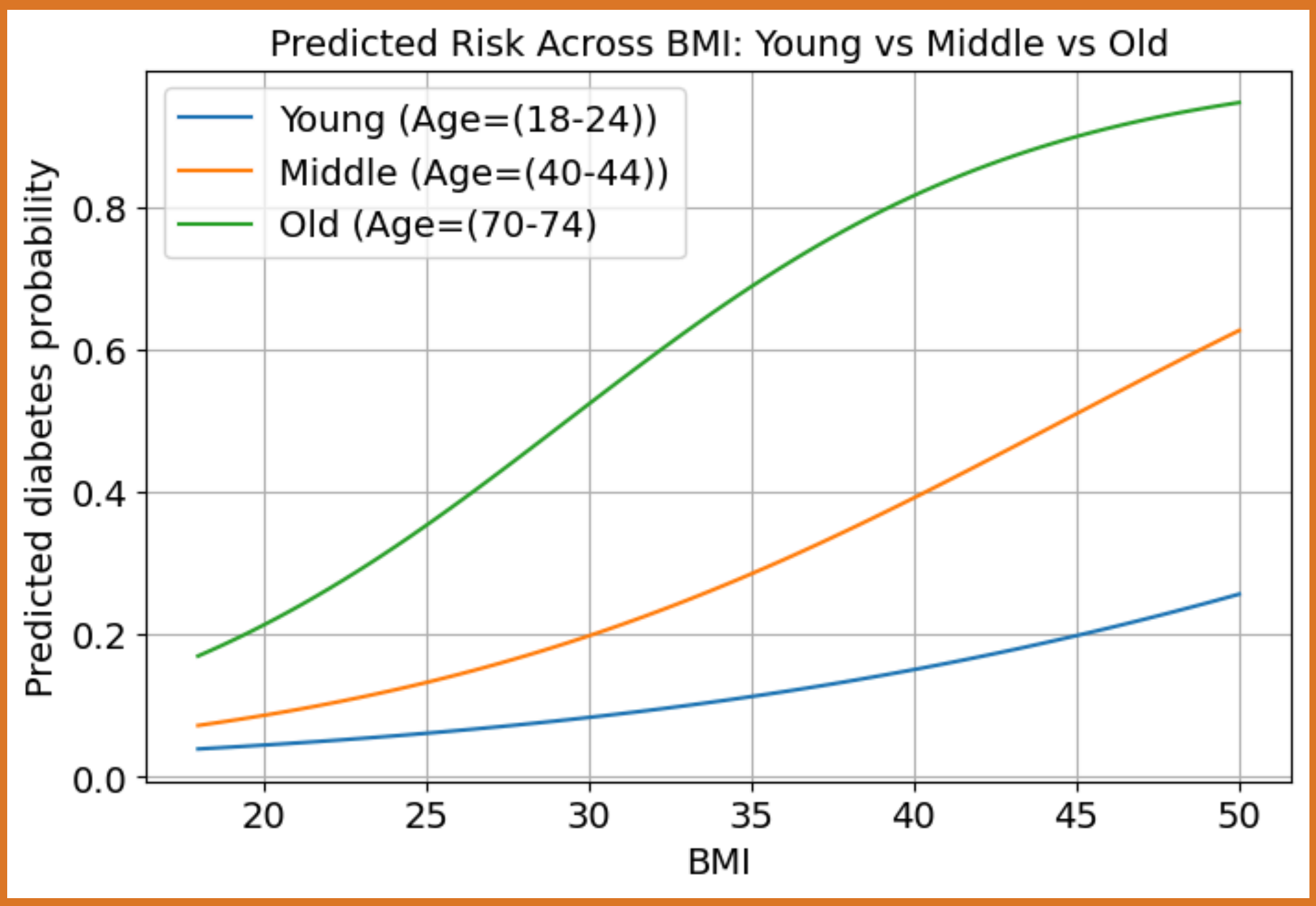

The x-axis shows range of BMI from 18 to 50. y-axis indicates predicted probability of having diabetes (0 - certainly no diabetes, 1 - certain diabetes)

This is a good example of how important it is to keep BMI low as we age. In the older age we can bring the almost certain chance of having diabetes to a very low one.

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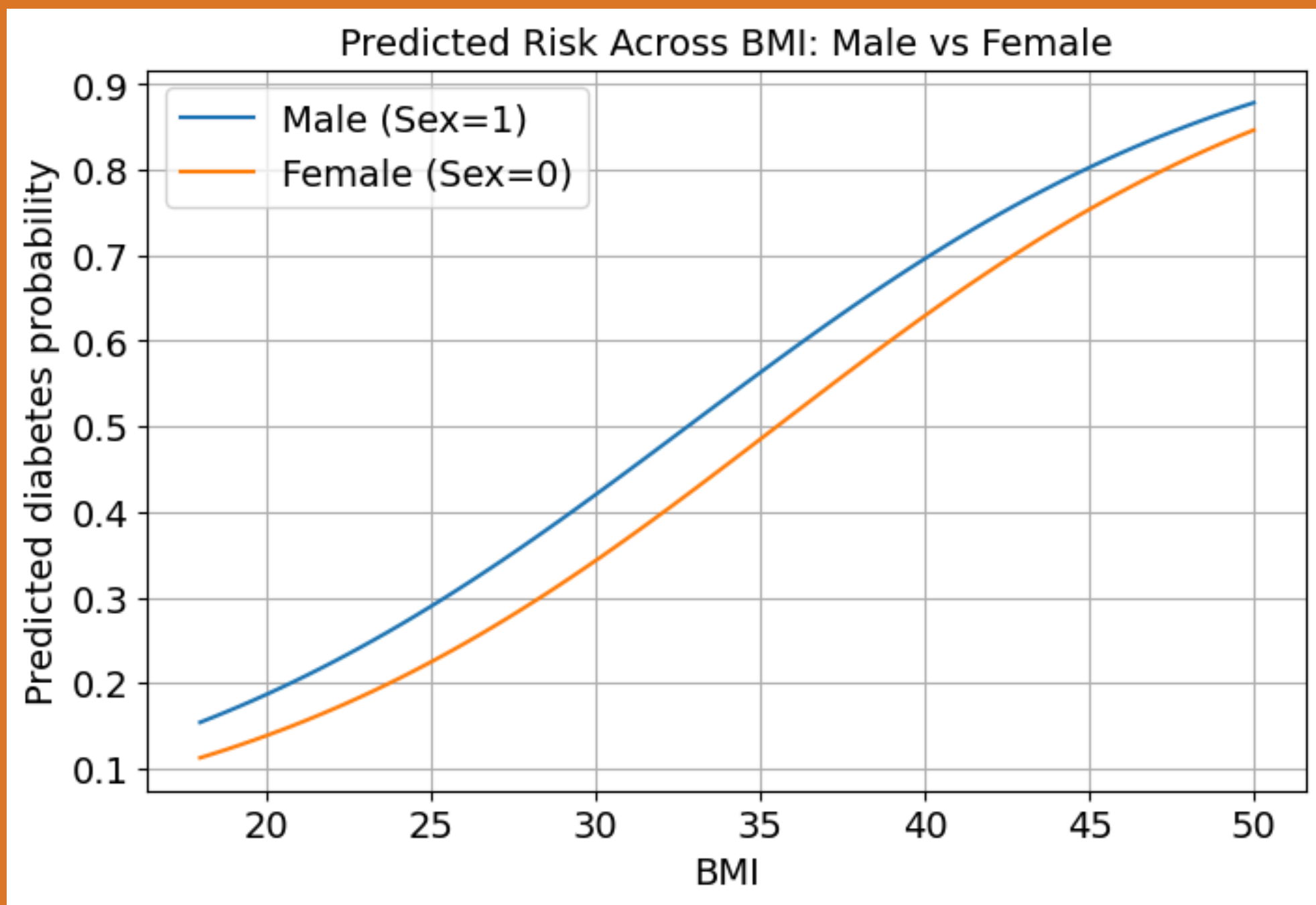

There is a small but consistent bias across all BMI values for males having higher probability of diabetes. We need to be aware of it.

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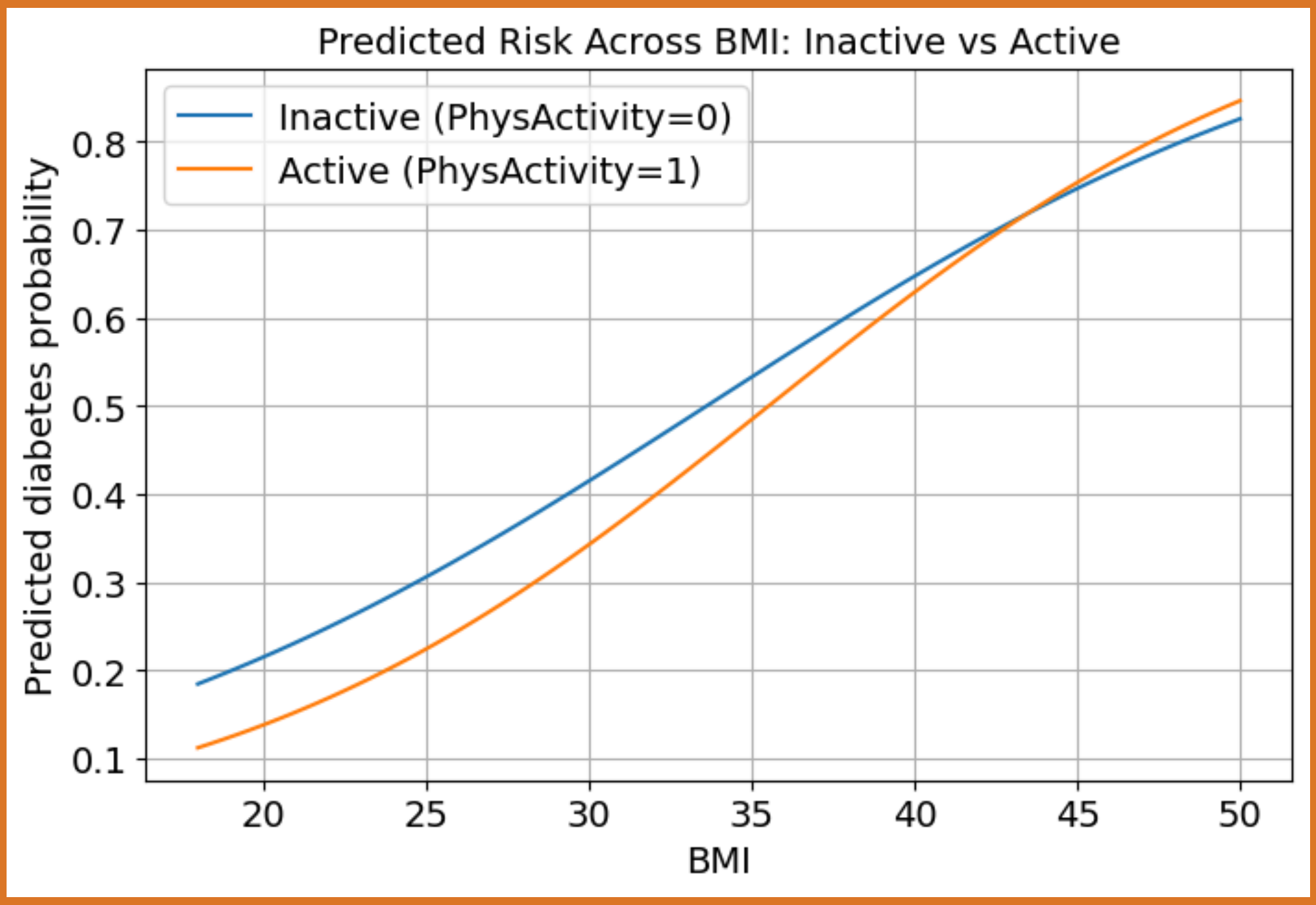

Another reason to keep BMI low: highest reduction of probability in diabetes thanks to exercise (almost 10%) only possible at low BMI. The advantage of exercise completely negates at BMI > 40

Conclusion

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BMI: lower BMI is especially critical as we age. Also exercise is most effective in reducing diabetes at lower BMI

Age: largest contributor , doubles chance of having diabetes every 4 years

Heavy alcohol consumption: Reduces chance of diabetes, this is probably reverse causality (diabetics stop drinking) - further investigation required

no health insurance and cant afford doctor: increases risk of diabetes by 20%

being in a bad physical shape: can increase chance of diabetes by 20%

gender female: females have 40% lower chance of having diabetes then males

income : individuals with higher incomes have at least 20% less chance to be affected by diabetes

Strongest positive effect due to exercise achieved only at lower BMI.

Keeping BMI low becomes critical as we age: large change in BMI can increase chance of diabetes by 10% in your 20s ...by 75% if you are in your 70s.

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